Project Initialization and Planning Phase

Date	23-June-2026
Team ID	SWTID-2026-7037
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To revolutionize crop selection for farmers by applying machine learning techniques that analyze soil nutrients and climate parameters, delivering fast and accurate crop recommendations.
Scope	The project comprehensively collects, analyzes, and models agricultural data (soil and environmental parameters) to build an MLbased crop recommendation system, deployed as a web application for real-time use by farmers and agricultural stakeholders.
Problem Statement	
Description	Farmers often struggle to determine the most suitable crop based on soil nutrients and climate conditions, leading to poor crop selection, inefficient resource utilization, and reduced agricultural productivity
Impact	Solving this improves crop yield, optimizes usage of water/fertilizers/soil nutrients, reduces financial losses from poor crop choices, and supports sustainable, data-driven farming decisions.

Proposed Solution	
Approach	Employ machine learning algorithms (Logistic Regression, KNN, Decision Tree, Random Forest, K-Means Clustering) to analyze soil and climate data and predict the most suitable crop, delivered via a Flask-based web application
Key Features	Implementation of an ML-based crop recommendation model trained on agricultural datasets

	Real-time crop prediction based on user-provided soil/climate inputs • Data preprocessing, feature scaling, and model evaluation for improved accuracy

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications,	Standard CPU (no GPU required — lightweight ML models)
Memory	RAM specifications	8GB
Storage	Disk space for data, models, and logs	10 GB (dataset + model files)
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn

Development Environment	IDE	Anaconda Navigator, PyCharm
Data		
Data	Source, size, format	Kaggle dataset (Crop_recommendation.csv), ~2200 rows, CSV format